

Project Pandora — Technical Design Specification

Product name: Turtle | Build codename: Project Pandora | Pilot: Hyderabad, iOS-first

REVISED THIS UPDATE — In-app currency renamed from Zar to Coins (placeholder pending a final name). Tile state machine updated (OTR renamed Activated, now dwell-progress-bearing, hybrid Unlock path); Activity Card system added in full; category taxonomy added; tech stack corrected to match locked V0 decisions (was stale); new V0 scope section added.

1. Purpose & Scope

This document specifies the technical architecture for Turtle's pilot build: a fog-of-war exploration app that rewards real-world movement, launching in Hyderabad on iOS. Companion document: "Turtle — Product & Strategy Overview."

Section 0A below defines what the V0 pilot build actually ships with, which is narrower than the full system this document specifies. Sections describing Quests, Creator compensation, Coins, and the full leveling framework (5–7, 10) remain the long-term architecture; treat them as the destination, not the pilot's starting point.

0A. V0 Pilot Build Scope

In scope

- Hex-grid fog rendering; tile states Unclaimed → Activated → Unlocked (see revised Section 3.2).
- Automatic tile clearing via GPS movement, foot or vehicle.
- Cumulative dwell-time progress and Activity Card boost, both feeding the same Unlock trigger (Section 3.2).
- Activity Cards: capture, categorization, map presentation (Section 3.5, new).
- Visual progress fill, streak counter, cohort leaderboard as the incentive layer.
- Deliberate, user-initiated background-location upgrade with persistent status indicator and graceful degrade.

Explicitly deferred

- Quest system (Section 5), Creator role and verification, CreatorScore, Coins (Section 6), Creator Compensation (Section 7).
- Squad tiles (Section 4.4).
- 100-level / six-tier Leveling Framework (Section 10) and its curve constants.
- Activity Card editing, carousel presentation, sharing, structured itinerary/cost/booking fields, multi-stop route visualization.

2. System Overview

Core loop: a user's real-world location clears fog over a hexagonal (H3) grid, revealing territory as they move. Users see only Area-level tiles and above — hex cells are an invisible backend signal. In the long-term design, progress is driven by Quests (authored by a small, vetted Creator population and taken by Explorers), a 100-level XP system, and Coins — non-transactable points earned through activity and redeemable only in Turtle's own catalog. Creators are compensated through a separate, off-ledger track, distinct from Coins. In V0, Quests and Coins are not live — see Section 0A.

3. Grid & Fog System

3.1 Grid technology & resolution

Built on H3, a hierarchical hexagonal spatial index, using standard resolution levels rather than custom cell sizing. Point-to-cell resolution is a pure function of latitude/longitude, computed identically anywhere on Earth with no database lookup. H3's native hierarchy (16 resolution levels, each parent composed of ~7 child cells) is the basis for the Area → District → City → Region progression tiers.

Pilot default: H3 resolution 9 (~174m average edge), chosen as the closest standard resolution to the ~100m radius reliable native geofencing can deliver.

Named-area boundaries (e.g. Begumpet, Prakash Nagar) are handled separately from the H3 tier hierarchy above — see Section 3.6, new.

3.2 Tile states — REVISED

State	Trigger
Unclaimed	Default, no data shown.
Activated (renamed from On the Radar / OTR)	Adjacent to an Unlocked cell, passed through, or entered with any dwell. Carries a hidden, cumulative dwell-progress value — NOT shown on the map by default; revealed on tap. This differs from the original OTR definition, which was non-reward-bearing; Activated now directly progresses toward Unlocked.
Unlocked	Reached via EITHER path, whichever comes first: (a) cumulative dwell-time threshold met, pilot-launch guess 10 minutes, stored as a server-side config value, never hardcoded; or (b) sufficient Activity Card boost (flat amount per tile touched by a logged Activity — see 3.5). In V0, with no Quest content live, dwell-time and Activity boost are the only two paths; the original quest-driven Unlock trigger becomes a third path once Quests ship post-pilot.
Mastered	See Section 4.3. Unchanged.

Dwell time is cumulative across separate visits to the same tile, not reset when the user leaves. Once Unlocked, a tile's state is never revoked or downgraded, including if the underlying threshold or Activity-boost constant is later retuned — compute current progress live from immutable per-user hex-clearance records; never store tile- or area-level totals as an independent stored fact.

3.3 Indoor / GPS-dead-zone handling

On GPS signal loss, the client freezes the user's last known outdoor position and pauses dwell-time accumulation until signal returns. True indoor positioning (BLE beacon) is reserved for the Event Quest vertical, tied to specific paying venues — not general city coverage.

3.4 GPS smoothing

Urban-canyon signal drift is handled via a rolling average over recent readings combined with implausible-speed filtering to reject bad pings. Elevation is not modeled — an accepted, permanent limitation. Quest difficulty is always a declared attribute, never sensor-derived, specifically to preserve this.

V0 location strategy: default “While Using” authorization; a deliberate, user-initiated action (“Turn on Live Fog Clearing”) triggers the real iOS system prompt to upgrade to “Always” — never requested passively. A persistent ON/OFF status indicator is shown whenever tracking is active. If “Always” is later revoked from iOS Settings, the app must detect this and degrade gracefully back to While-Using behavior. Tile updates use significant-location-change rather than continuous background tracking, arriving in batches — UI is designed around “fog catching up,” with a manual foreground refresh available, rather than promising real-time clearing.

3.5 Activity Cards — NEW

An Activity Card is an independent entity, not tied to a single tile — it is a multi-tile summary of an outing, event, or day. A user taps “Start an activity,” optionally names it, and captures one or more photos or voice notes as they move; each capture is tagged with tile and timestamp. Tapping “End” closes the session. A single opportunistic capture with no explicit start/end still creates a minimal one-tile Activity. An open session auto-ends after roughly two hours of inactivity to avoid orphaned sessions absorbing unrelated later captures.

Progress boost: every tile touched during a logged Activity receives the same flat boost toward Unlocked — no weighting by per-tile dwell time in V0. This is a placeholder constant, tuned the same way as the dwell threshold (Section 3.2), via server-side config.

Map representation: one pin per Activity, placed at its start location (or centroid), color/icon-coded by category (Section 3.6). Tapping the pin reveals the tiles and category covered. Multi-stop route or connected-pin visualization is explicitly out of scope for V0 — that belongs to the deferred itinerary/sharing feature set.

V0 Activity Cards carry no editing, no carousel presentation, no sharing, and no structured itinerary/cost/booking fields — those are Phase 2+. Post-pilot, Activities are a natural pipeline for crowdsourced Quest candidates once a Creator population exists.

3.6 Activity Categories — NEW

Fixed list for V0, stored in an admin-editable Supabase table (not hardcoded, not user-extensible) so it can expand without a rebuild. “Explorer” remains reserved exclusively for the user-role name and is never reused as a category-system name.

Category	Subcategories
Sightseeing & Heritage	Monument/Fort, Museum, Religious site, Heritage walk
Dining	Restaurant, Cafe, Street food/stall, Food truck
Nature & Outdoors	Park, Lake/waterfront, Garden, Viewpoint
Sports & Adventure	Ground/stadium, Gym/fitness, Adventure activity, Water sports
Events	Concert/gig, Exhibition, Festival, Pop-up/market
Nightlife	Bar/pub, Club, Lounge
Wellness	Spa, Salon, Yoga/meditation
Travel	Scenic drive, Night drive, Bike ride, Walk, Trek, Cycling

Category is selected once per Activity, at logging/end time rather than upfront, to minimize friction at the moment of capture.

3.7 Named-area boundaries — NEW

Named localities (e.g. Begumpet, Prakash Nagar, Brahmanwadi) are handled as a one-time offline mapping, separate from the live H3 tier hierarchy in 3.1: source boundary polygons from OSM and/or GHMC ward data, verified per-locality since coverage is inconsistent (some localities exist only as an address-level point with no traced polygon), then compute which H3 cells fall inside each named area and load the result as a static config file (h3_id → area_name). No in-app boundary-editing tool is built for V0 — this is a one-time script, not a live admin feature.

Because this is a lookup table over immutable hex-clearance records rather than a stored per-user fact, subdividing an area later (e.g. splitting Begumpet into finer localities) requires no data migration and never revokes progress already computed against the coarser definition.

4. Territory Progression & Tile System

4.1 Base declaration

- A user must declare a Base (home city/town) before the rest of the app is usable. Declaration requires verified physical presence — no remote or self-reported declaration path.
- Re-declaration is always available, always presence-verified, gated by a 90-day cooldown, with reminders escalating in frequency as the cooldown nears its end (gentle at day 60, firmer at day 75, final notice at day 85–89).
- Base auto-unlocks the city tile to Unlocked and the state/region tile to Activated, with zero Coins reward attached.

4.2 Per-user, non-exclusive progression

Territory tiers are computed independently per user — there is no ownership contest between users. State is a live computation against current cell/tier definitions, not a stored claim, so administrative edits to boundary or naming data never require revoking anything from any user.

4.3 City/Area state, progress, and Mastery

- State is a binary threshold: a City flips to Unlocked the moment any one Area inside it reaches Unlocked.

- Progress is a continuous percentage, computed against underlying hex-cell coverage (not raw Area count) so cities with different admin-drawn Area counts remain comparable. Displayed in Low / Mid / High bands.
- A City or multi-tile Area reaches Mastered at 30% or greater hex-cell coverage. An atomic Area tile with no sub-tiles reaches Mastered once a user completes a minimum of 5 quests spanning 5 different Categories within it (long-term design, post-Quest-launch).

4.4 Squad tiles (deferred — see Section 0A)

Personal tile state is always individually earned through the user's own verified activity — Squad membership never changes a user's own state. A Squad is a tagging/overlay layer: a tile becomes Squad-tagged the moment any member personally unlocks it and chooses to share it. Users may belong to multiple Squads; each has its own badge. Sharing is opt-in and private by default.

4.5 Historical backfill

Deferred to a later phase. iOS provides no public API for third-party access to historical location data; any future version of this feature depends on photo-geotag extraction (inherently partial) or user-initiated manual import.

5. Quest System (long-term design, deferred — see Section 0A)

5.1 Type, Category, and Difficulty

Every quest is defined along three independent axes. Type is the verification mechanism; Category is the theme; Difficulty is relative to the user's current level tier, not an absolute scale.

Type (pilot)	Verification approach
Trivia	An on-site-only question with an exact or multiple-choice answer — no photo or AI review, resolves instantly.
Photography	A Creator-set reference image and framing; the Explorer captures a matching shot in-app, verified by AI content-matching plus presence metadata, with human review on low-confidence submissions.
Hidden Item	Same verification pipeline as Photography, applied to a specific real-world detail rather than a framed shot.

Additional lower-cost formats (a placed physical code to find and enter, a simple timed check-in, a multiple-choice “spot the real one,” and a participation-based sketch/creative submission) extend pilot content without requiring new verification infrastructure. Categories are open-ended — History, Food, Architecture, Drive-by, People, and others.

5.2 Quest state machine

State	Trigger / Effect
Start	Explorer accepts; timer begins. No reward at this point.
Alert	No progress within the first third of the time limit.
Reset	No response to the alert, or the timer expires with presence never verified. No unlock, no reward to anyone — distinct from Failure.
Failure	Presence was verified but the task wasn't completed or didn't pass review. The tile still Unlocks — genuine effort is rewarded with permanent progress even without points.
Success	Task completed in time, submitted, and passes review. Tile Unlocks and the Explorer earns the full point reward for the quest.

5.3 Quest Card lifecycle

A Quest Card is the content object a Creator authors — reference material, Type, Category, and Difficulty tag — with a default six-month lifetime and unlimited reuse across Explorers during that window. Community upvoting (1,000 upvotes) unlocks, but does not automatically trigger, a Creator-controlled option to extend a quest's lifetime by roughly a month per threshold reached.

5.4 Meta Quests

System- or admin-authored quests with no Creator, no purchase, and no timer. ‘Unlock Tiles’ is the default onboarding Meta Quest. Event Quests (sponsored live events, with a venue-wide micro-map) are a validated future direction, not part of the pilot build.

5.5 Creator / Explorer roles

Creator status is earned, not self-selected. All non-employee paths require a background check via a licensed verification provider. Every Creator-authored quest also undergoes its own safety review, independent of the Creator's own vetting. Photo/video submissions are captured only through Tourtle's own in-app camera, never a gallery upload, to preserve trustworthy location and timestamp metadata.

5.6 Community layer

Quest cards support ratings, reviews, comments, likes, shares, and tagging. Explorers can follow Creators, whose new quests then surface on the follower's map with a notification.

6. Coins — Explorer Points (long-term design, deferred — see Section 0A)

Coins are non-transactable points: earned through verified activity and spendable only in Tourtle's own redemption catalog. No peer-to-peer transfer, no payment to Creators in Coins, no purchase with real currency, no cash-out.

Source	Coins	Notes
Check-in at a real location	5	Requires an actual establishment; limited by a daily per-location cooldown
Tile Unlock	10	Same event whether triggered passively or through a quest
Quest success	Scales with Difficulty	Higher-difficulty quests pay more
Quest failure	0	Tile still Unlocks — the non-monetary reward is preserved
City Mastered (Meta Quest)	100	—

Anti-farming protections include a daily per-user-per-location check-in cooldown, device fingerprinting, phone verification, account-creation velocity limits, and dwell-time-verified presence for every reward-bearing event.

7. Creator Compensation (long-term design, deferred — see Section 0A)

Creators are compensated separately from the Coins system. At pilot stage, compensation consists of status and recognition, access and privilege, real-world promotional exposure, and a banked performance score tracked from day one for retroactive payout once Tourtle reaches a funded stage.

CreatorScore = weighted sum of: verified completions + average star rating + distinct-Explorer reach + footfall-to-completion ratio. Scored on a trailing 30-day rolling basis. Once real revenue exists, compensation transitions to a genuine revenue-share Creator Fund, paid monthly through standard banking rails with applicable Indian tax withholding.

8. Privacy Architecture

Raw exploration data is local-first by default; sharing to the server, including to a Squad, is always an explicit, per-user opt-in action.

VO addition: location, camera, and microphone all require clear, specific purpose strings and an explicit consent screen before first use, in line with active DPDP Act obligations — not deferred to any later enforcement milestone.

9. Anti-Fraud & Device Integrity

- Device fingerprinting, phone verification, account-creation velocity limits, and cross-account device correlation guard against multi-accounting.

- Mock-location and GPS-spoof detection runs at every point-earning event and gates Base declaration and re-declaration.
- Creator-side gaming is guarded at the CreatorScore level — device-linked completions and low-tenure accounts are excluded from contributing to a Creator's score.

10. Leveling Framework (long-term design, deferred — see Section 0A)

Tier	Levels	Pacing	Feel
Spawn Point	1–10	Fastest, first level same session	Tutorial-adjacent
Grid Starter	11–25	Fast, low friction	Habit forming
Zone Runner	26–40	Mild slowdown	Established casual
Grid Vet	41–60	Notably steeper	Real commitment; Creator eligibility opens around Level 45
Uncharted	61–90	Steeper, longer gaps	Serious dedication
Grid Master	91–100	Steepest, deliberate grind	Endgame status

XP is a separate, cumulative, never-decreasing value, independent of Coins balance. The curve is calibrated toward a casual user reaching Level 10 in about a month and a dedicated user reaching Level 100 in roughly two years.

11. Event Quests (Future Vertical)

A validated future opportunity: sponsored live events as venue-wide Meta Quests, using BLE beacon positioning. The intended approach is to license existing indoor-positioning infrastructure rather than build it in-house. Sequenced after the core pilot stabilizes.

12. Tech Stack — CORRECTED

Layer	Choice & rationale
Client (V0 pilot)	Flutter (locked, not “RN or Flutter” as previously listed) — native modules for background geofencing specifically.
Maps / fog rendering	MapLibre GL, not Mapbox as previously listed — avoids per-load metered billing exposure, appropriate for a self-funded solo pilot.
Grid engine	H3 (client + server), resolution 9.
Backend / data	Supabase, not raw PostgreSQL as previously listed — Supabase runs on Postgres underneath, so transactional integrity for any future Coins/CreatorScore ledger work is preserved, while giving a solo founder auth, storage, and realtime out of the box.
Location	Platform-native geofencing (significant-location-change for V0) with a last-known-fix fallback for signal loss.
Anti-fraud	Apple device attestation and Play Integrity API, alongside device-fingerprinting and velocity-limit services (post-V0).
Identity/background checks	A licensed India-based verification vendor (post-V0, once Creator role is built).
Analytics	Live from launch, feeding dwell-threshold and Activity-boost constant tuning in V0, and the leveling/quest curves post-launch.

13. Outstanding Items for Future Resolution

- Final XP curve constants and Meta Quest scoring curve, pending real pilot telemetry.

- Difficulty-per-level-band Creator guidance content.
- Creator Fund scaling policy and minimum payout threshold for Phase 2.
- Final disclosure language for the banked CreatorScore commitment.
- Footfall-vs-completion popularity analytics pipeline.
- Community-layer and content-review moderation staffing plan.
- Quest Card expiry grace handling for in-flight attempts.
- V0 dwell-time threshold (10 min guess) and Activity-boost flat amount, pending real pilot telemetry — both stored as server-config, not hardcoded.
- Whether the background/Always location entitlement functions correctly under a free personal-team Apple account — unverified, to be confirmed in Phase 0 of the build.
- Apple Developer Program enrollment funding (\$99/yr) — hard gate on any external pilot distribution.
- Per-locality OSM/GHMC boundary polygon verification for Hyderabad sub-localities lacking a traced shape.